

Chapter 21

Syncing Audio and Video

When you're working on a program where the production audio was recorded separately from the production video (often referred to as "dual-system recording"), DaVinci Resolve provides tools for syncing the audio and video together in a variety of ways to create media that you can edit easily. The process of syncing audio and video together is often referred to as "syncing dailies."

Contents

Syncing Audio to Video.....	438
Syncing Audio to Video Using Timecode.....	438
Syncing Audio to Video by Matching Waveforms.....	439
Manually Syncing Audio to Video.....	440
Retaining Video Metadata and Embedded Audio when Syncing Audio Manually.....	441
Offsetting the Sync of Previously Synced Clips.....	442
Finding Synced Audio Files.....	442
Displaying Synced Audio File Names on the Timeline.....	443

Syncing Audio to Video

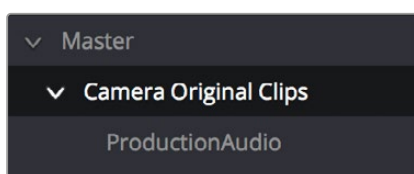
If you're processing dailies for a shoot that used dual-system recording, where audio is recorded to a separate device than video, you can "sync the dailies" in DaVinci Resolve in one of two ways. Synced clips can be output as media files with embedded audio or output to tape, whatever your client requires.

Syncing Audio to Video Using Timecode

Ideally, if the sound recordist on set was highly organized, and the camera and audio recorder both used synchronized timecode, you can use a single command to automatically sync every clip in a timeline to a bin of Broadcast .wav files that have matching timecode.

To batch sync audio to video using timecode:

- 1 Create a new project, open to the Media page by default, and import the video media you need to sync into any bin of the Media Pool.
- 2 Import the matching Broadcast .wav files into the same bin as the accompanying video media you imported in Step 1. If you want to stay more organized, you can create another bin to contain the audio clips, but it must be inside the bin that contains the video files. The audio bin can be named anything you like.



Organizing production audio in a bin created within the accompanying camera media bin

- 3 Right-click the bin containing the matching audio and video clips, and choose one of the following commands from the contextual menu:

Auto-Sync Audio Based on Timecode: Replaces each video clip's previous audio channels with audio channels from the newly synced .wav files.

Based on Waveform: Analyzes and compares the waveforms of each of the selected clips and replaces each video clip's previous audio channels with the newly synced .wav files.

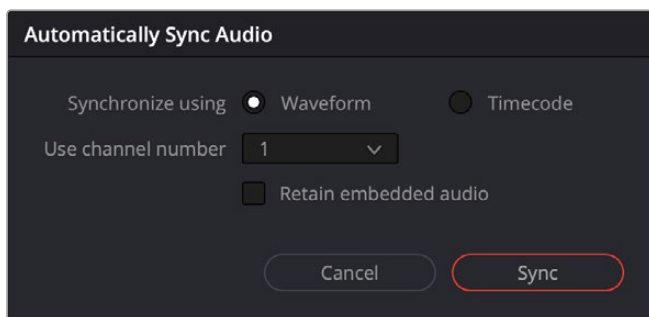
Retain embedded audio: Adds new channels in addition to the audio channels that were previously in the media file. The newly waveform-synced channels are added to an additional track, so when edited into the timeline, a clip that's synced this way appears with one video clip and two audio clips that occupy two different audio tracks, so you can edit the camera original audio independently from the synced audio.

Every clip in the selected bin for which there was an accompanying Broadcast .wav file with matching timecode is immediately synced with an audio track. If multiple audio files overlap with matching time code, each file will be synced and a new audio track added to the resulting clip to accommodate each audio file. You can modify this behavior to only sync the single best matching file and ignore others by checking the "Limit media pool audio sync to first timecode match" box in the Editing panel of the User section in the DaVinci Resolve Preferences.

All synced clips appear with an audio icon at the bottom left in the Media Pool when Thumbnail view is selected. Now that the clips are synced, you can edit them in the Edit page or use the Deliver page to export offline dailies or online media with embedded sync audio for use in other applications.

Syncing Audio to Video by Matching Waveforms

If you don't have matching timecode in the audio and video source clips you're syncing, but you had the foresight to record camera audio at the same time as the dual source production audio you want to sync to, DaVinci Resolve can use waveform syncing to compare the audio waveforms of your audio and video source files, and sync the ones that match.



Progress dialog for syncing dialog using waveforms

To batch sync dailies using waveform syncing:

- 1 Create a new project, open to the Media page by default, and import both the video and audio media you need to sync. There's no need to organize your files in any particular way, but it's not a bad idea, on multi-day shoots, to organize the audio and video files so that it's easy to select all of a single day's clips at once so that you can sync your files in smaller batches. Even organizing your clips by scene can make waveform syncing go faster by reducing the number of files that need to be compared at once.
- 2 If you've placed the audio and video into separate bins, then you can Command-click both bins in the bin list to select them and expose all of their contents in the Media Pool. If you placed your media in the same bin, this is not necessary.
- 3 Select one of the exposed clips in the Media Pool, and press Command-A to select all audio and video clips you want to sync.
- 4 Right-click one of the selected clips, choose Auto Sync Audio from the contextual menu, and select one of the methods below.

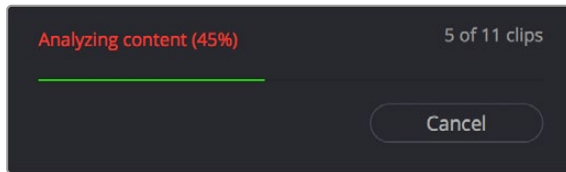
Based on Timecode: Synchronizes the timecode between the audio and video clips, and replaces each video clip's previous audio channels with the newly synced .wav files.

Based on Waveform: Analyzes and compares the waveforms of each of the selected clips, and replaces each video clip's previous audio channels with the newly synced .wav files.

Use channel number: Lets you choose which audio track to perform the waveform analysis on. With the default of "Auto," DaVinci Resolve automatically picks the channels most suited for waveform comparison. With "Mix," DaVinci Resolve compares waveforms from the mixdown audio from all available channels. This can be useful when audio may not be present on long sections of one or channels. Users can also specify individual channels to avoid noisy audio.

Retain embedded audio: Analyzes and compares the waveforms of each of the selected clips, and adds new channels in addition to the audio channels that were previously in the media file. The newly waveform-synced channels are added to an additional track, so when edited into the Timeline, a clip that's synced this way appears with one video clip and two audio clips that occupy two different audio tracks, so you can edit the camera original audio independently from the synced audio.

A progress bar dialog appears, showing you how long the syncing operation will take. When it's complete, your clips will be synced.



Progress dialog for syncing dialog using waveforms

TIP: After syncing, you may be notified via a dialog that one or more clips could not be synced. Note these clips, as it may be possible to use waveform syncing more successfully on just the selected pair of audio and video items that belong together.

Manually Syncing Audio to Video

If you have a collection of WAV or AIFF audio files with video source media that lacks matching timecode, you need to manually sync each pair of media files together, one-by-one, using a sync reference such as the clap of a clapperboard or any other sharp sound with a distinct audio/visual correspondence.

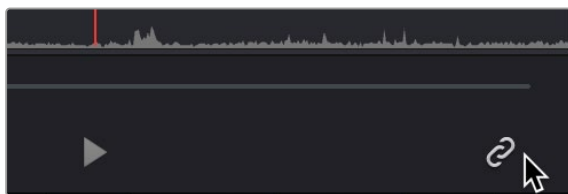
To manually sync audio to video:

- 1 Create a new project, and import the video media you need to sync into the Media Pool. If a dialog appears asking whether or not you want to update the project to match the media, click OK.
- 2 If you want to stay organized, create a second bin in the Media Pool, named Audio Clips, and import the matching Broadcast .wav files into it. The name of the bin is not important, and having all the audio in one bin is simply a matter of convenience.
- 3 Click the Waveform button at the top of the Audio Panel, which lets you view and scrub along the waveform of audio clips you select in the Media Pool.
- 4 Select a video clip to sync, and move the Viewer playhead to line up with the first visual sync point in the first clip. This could be the clap of a clapperboard, the red flash of a tablet computer's slate app, a hand clap, or any clear visual cue to which there is a corresponding audible sound.
- 5 Now, select whichever audio clip corresponds to the current video clip in the Viewer, to open its waveform into the Audio Panel.
- 6 Use the Audio Panel transport controls and scrubber bar in the Source Viewer to move the playhead to the audio sync point that corresponds to that video sync point. This may be a clap, a beep, or some other staccato sound that's easy to sync to. As you play through the clip, the bottom half of the Viewer shows a zoomed out waveform for the entire clip, while the top half of the Viewer shows a zoomed in section of the waveform that immediately surrounds the playhead. Hopefully, the sync point you're looking for is a distinct, loud spike somewhere towards the beginning or the end (in the case of a tail slate) of the audio clip.



Aligning video and audio sync points with the Audio Panel set to show the Waveform panel

- 7 When you've found the audio sync point that matches the video sync point, click the Link/Unlink Audio button at the bottom right of the Audio Panel to embed the now synced audio into the video clip.



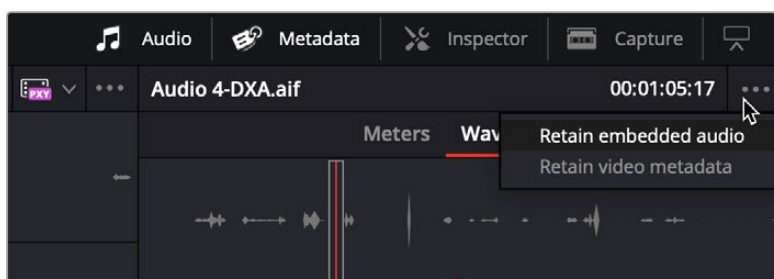
Clicking the sync link button to lock sync

The audio and video items are linked. At this point, you can use the newly synced clips in the Edit page, and use the Deliver page to export offline or online media with embedded audio for editing.

Retaining Video Metadata and Embedded Audio when Syncing Audio Manually

When syncing audio to video manually in the Media page, you can choose whether to include the original camera audio tracks in the synced file, or whether to retain the camera scene/take metadata, rather than have the audio file scene/take metadata overwrite it as normal.

In the Media page > Audio > Waveform tab, the option (3 dot) menu now offers 2 choices:



The Audio > Waveform tab (3 dot) option menu lets you choose to retain embedded audio and/or video metadata.

Retain embedded audio

- When enabled, embedded camera audio tracks are included with any newly synced audio during the manual linking process (using the link icon on this pane).

Retain video metadata

- Default is off, which means that video files that are synced to audio files inherit metadata (e.g., scene and take) from the audio file. When this option is checked, the video file's metadata is retained and not overwritten instead.

Offsetting the Sync of Previously Synced Clips

If you need to offset the audio (or stereo 3D) sync of the items that make up a clip later on, you need only select the synced clip you want to resync in the Media Pool, then click the Waveform button at the top of the Audio panel to show the clip's audio waveform, turn off the linked clip button, change either the audio or video sync points, and turn the linked clip button back on again.

You can also use two sets of commands for slipping the sync of any clip:

- **Trim > Slip Audio > Slip Audio One Frame Forward/Reverse:** (Option-Period and Option-Comma) Slips the audio/video sync of any clip in whole frame increments.
- **Trim > Slip Audio > Slip Audio One Subframe Forward/Reverse:** (Option-Right Arrow and Option-Left Arrow) Slips the audio/video sync of any clip in 1/10th frame increments.
- **Trim > Slip Eye > Slip Eye One Frame Forward/Reverse:** (Command-Option-Period and Command-Option-Comma) Slips the sync relationship between the eyes within a stereo clip in whole frame increments.

Finding Synced Audio Files

When you've synced dual-system audio and video clips together in DaVinci Resolve, you can find the audio clip that a video clip has been synced to using the following procedure.

To find the audio clip that a video clip has been synced to:

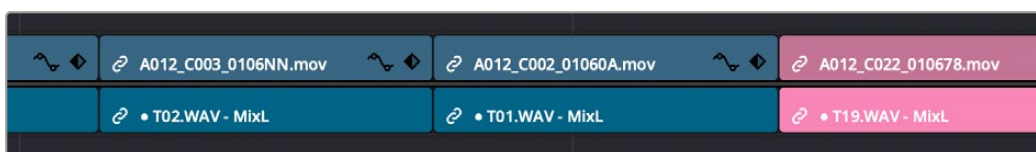
- Show the Media Pool in List view, and reference file name in the Synced Audio column.
- Right-click a video clip that's been synced to audio, and choose "Reveal synced audio in Media Pool" from the contextual menu. The bin holding the synced audio clip is opened and that clip is selected.

Displaying Synced Audio File Names on the Timeline

For certain workflows you may wish to see the name of the original audio file used in a synced dual system audio pair on the timeline tracks, rather than the name of the video clip its attached to.

To display the filename of the original audio file used in a synced pair in the timeline:

- 1 Choose View > Show File Names. You cannot see synced audio file names unless you've set DaVinci Resolve to display the original file names.
- 2 Choose View > Overlay Synced Audio File Names. You should now see the names of the synced audio files superimposed on the audio clips in the Timelines, and the names of the video files superimposed on the video clips in the Timelines, even when they're synced.



Viewing the synced audio file names in the Edit page Timeline